# 有效介入預防疾病
# 同步口譯準備教材

*Effective Interventions That Prevent Disease*

**170 萬**

每年 5 歲以下兒童死亡歸因於環境 / WHO 2017

## SPEC

| | |
|---|---|
| 研討會 | 2017 亞太兒童環境健康研討會 Asia-Pacific Children's Health Symposium｜Day 1 |
| 時間地點 | 2017/11/08（三）14:00｜長榮國際會議中心 1101 室 |
| 場次 | Session I：Lifestage Approach and Intervention｜主持：鄧振中（北醫講座教授） |
| 講者 | Dr. Ruth A. Etzel（MD, PhD），美國環保署兒童健康保護辦公室（OCHP）；口譯稱「Etzel 博士」 |
| 同場 | 13:30 Walter Rogan〈Nine Months that Last a Lifetime〉；14:30 座談 |

## CONTENTS


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## COLOR KEY

| | |
|---|---|
| 術語底線 | #78A9D6 |
| 知識點 | #2B7A78 |
| 口譯技巧 | #B0461E |
| 投影片 | #1F3A5F |
| 數字預讀 | #8A6212 |

# 全場地圖

SPEECH MAP

00 地圖

7 段 / 朗讀約 4-5 分鐘 / 34 個數字

講者的論證順序，一段一格。箭頭旁是講者自己的轉折句：聽到它，下一段就開始了。




各段長度依字數比例，以每分鐘 130 words估算

## 01 開場：兒童不是縮小版的大人

0:00

兒童環境脆弱性是國際共識

TERMS *G8* / *children are not little adults*

FIGURES

| | |
|---|---|
| 1997 | 3,300 |

↓

## 02 兒童為什麼不同

0:34

暴露多、器官未熟、解毒弱、餘命長

TERMS *hand-to-mouth activity* / *metabolic rate* / *skin surface area* / *excretion*

↓

*Child **chronic diseases** and **developmental disabilities** are increasing.* 兒童慢性病和發展障礙正在增加。

## 03 疾病負擔正在上升

1:14

每年 170 萬五歲以下死亡歸因環境

TERMS *chronic diseases* / *developmental disabilities* / *desertification* / *deforestation*

FIGURES

| | |
|---|---|
| 1.7 million | 570,000 |

↓

## 04 戶外空汙：法規見效

1:42

《清淨空氣法》效益超過成本 30 倍

TERMS *Outdoor air pollution* / *asthma* / *Clean Air Act* / *Air Quality Index (AQI)*

FIGURES

| | |
|---|---|
| 1970 | $2 trillion |
| $65 billion | 230,000 |

↓

*Exposure to **biomass fuel** causes **household air pollution**.* 接觸生質燃料，會造成家戶空氣汙染。

## 05 家戶空汙：改良爐灶

2:20

兒童氣道窄，換乾淨爐灶報酬高

TERMS *biomass fuel* / *household air pollution* / *cleaner-burning stoves* / *return on investment*

↓

## 06 菸害：從魅力到反派

2:49

禁菸法規有效，但只涵蓋 18% 人口

TERMS *Secondhand smoke* / *Framework Convention on Tobacco Control (FCTC)* / *clean indoor air laws* / *preterm births*

FIGURES

| | |
|---|---|
| 600,000 | 1% |
| 1.8 billion | a quarter |

↓ *Lead exposure accounts for about 1% of the global burden of disease,* 鉛暴露約占全球疾病負擔的 1%，

## 07
3:40

### 鉛：汽油成功，油漆未竟
禁用含鉛汽油使血鉛下降，油漆仍待努力

TERMS *mental retardation* ／ *leaded gasoline* ／ *primordial prevention* ／ *Lead in paint*

FIGURES

| | |
|---|---|
| 1% | 20% |
| 1970s | 2010 |

# 背景解說
HOW IT ACTUALLY WORKS

3 節 / 每節附資料來源

01



講者沒說出口的因果機制。先讀懂這幾節，聽到相關段落時就能預判下一步。

## 01 預防金字塔：根本預防和初級預防差在哪？
THE PREVENTION PYRAMID: PRIMORDIAL VS PRIMARY

**Q.** 講者把含鉛汽油、菸草公約都叫 primordial prevention，為什麼不是 primary？

***Why does the speaker call the lead and tobacco measures primordial rather than primary prevention?***

| ENGLISH | 中文說明 |
|---|---|
| From the base up: **primordial prevention** keeps the risk factor from existing at all (banning leaded gasoline, setting air quality standards); **primary prevention** stops disease where the risk is already present (handwashing, cutting exposure); **secondary prevention** catches it early (blood lead screening); **tertiary prevention** treats disease that has occurred (chelation). | 四層由下而上：根本預防讓危險因子根本不存在（禁用含鉛汽油、立空氣品質標準）；初級預防在危險因子已存在時防止發病（洗手、減少暴露）；次級預防早期發現（血鉛篩檢）；三級預防治療已發生的疾病（螯合治療）。 |
| Her argument is that the lower you go, the more people you cover, the more you get per dollar, and the less you depend on individual behavior. Rogan's chelation trial shows that tertiary prevention can't undo damage already done. | 講者的論點是：越往下層，覆蓋人口越廣、每元效益越高，也不必依賴個人行為。Rogan 的螯合試驗正說明三級預防救不回已造成的傷害。 |

出現在
投影片第 28、49 張金字塔；第 7 段

可以預判
講者可能說 "move upstream"→「往上游走」「從源頭處理」。

SOURCES Etzel 投影片第 28、49 張；預防層級定義為公衛通用概念（已知內容）

## 02 《清淨空氣法》的成本效益怎麼算？
HOW THE CLEAN AIR ACT COST-BENEFIT IS COMPUTED

**Q.** 2 兆美元效益是怎麼來的？

***Where does the $2 trillion benefit come from?***

| ENGLISH | 中文說明 |
|---|---|
| Under Section 812 of the 1990 Amendments, EPA periodically estimates the Act's costs and benefits. Most of the benefit comes from **avoided premature deaths** and illness, converted to dollars with the value of a statistical life, which is why 230,000 avoided deaths dominate the total; the cost is what industry and government spend on **pollution control**. | EPA 依 1990 年修正案第 812 條規定，定期估算法規成本與效益。效益主要來自避免的早逝與疾病，用「統計生命價值」換算成金錢，所以避免 23 萬人早逝就占了大部分效益；成本是產業與政府的汙染控制支出。 |

ENGLISH

The AQI is the tool for individuals: it runs from 0 to 500, 100 roughly equals the national standard, and above 100 the air is unhealthy for sensitive groups, children included. Taiwan also uses the AQI.

中文說明

AQI 則是給個人用的工具：0–500，100 約等於國家標準，超過 100 對敏感族群（包括兒童）不健康。台灣環境部也採用 AQI。



出現在

投影片第 24–27 張；第 4 段

可以預判

"value of a statistical life"→「統計生命價值」。

SOURCES US EPA, The Benefits and Costs of the Clean Air Act from 1990 to 2020 (2011)（投影片備註引用）；EPA AirNow AQI 說明（投影片備註）

## 03 含鉛汽油現在還有人用嗎？
IS LEADED GASOLINE STILL SOLD ANYWHERE?

**Q.** 投影片地圖截至 **2016** 年，現在情況如何？

***The slide map is as of 2016. What is the situation now?***

ENGLISH

Tetraethyl lead was added to gasoline to stop engine knock; it left the exhaust as dust in air and soil, which children inhaled or swallowed through hand-to-mouth activity. Wherever it was phased out, blood lead fell sharply. In 2016 a few countries such as Algeria, Iraq and Yemen still used it; after **Algeria stopped selling it in 2021**, UNEP declared leaded petrol for cars eliminated worldwide.

中文說明

四乙基鉛當初加入汽油是為了抗爆震，鉛經排氣散布到空氣和土壤，兒童吸入、經手口行為吃下。各國淘汰後血鉛普遍大幅下降。2016 年時仍有阿爾及利亞、伊拉克、葉門等少數國家使用；阿爾及利亞 **2021** 年停售後，UNEP 宣布全球道路車用含鉛汽油已全面淘汰。

出現在

投影片第 44–46 張；第 7 段

可以預判

如講者在座談更新此事，可說「2021 年全球已全面淘汰」。

SOURCES UNEP 新聞稿 2021/08/30〈Era of leaded petrol over〉（已知內容，未重新查證網頁）

# 術語解說
TERMINOLOGY

02

59 個術語 / 5 類

依素材順序歸類。原文以粗體色字標示，建議譯法以黑體標示，多個譯法以／分隔。



MAIN TERMS
## 核心術語 6 個；格子標出它在講稿第幾段出現

**E** 預防層級 — **2** 次
1 2 3 4 5 6 **7**
**primordial prevention**
根本預防
從源頭不讓危險因子出現，如禁用含鉛汽油

**D** 汙染源與介入 — **2** 次
1 2 3 **4** 5 6 7
**Air Quality Index**
(AQI)
空氣品質指標（**AQI**）
台灣同用 AQI

**D** 汙染源與介入 — **2** 次
1 2 3 **4** 5 6 7
**particulate matter**
(PM)
懸浮微粒

**A** 機構、法規與文件 — **2** 次
1 2 3 4 5 **6** 7
**Framework Convention on Tobacco Control**
(FCTC)
《菸草控制框架公約》
2003 年通過

**C** 疾病與負擔 — **2** 次
1 2 3 **4** 5 **6** 7
**asthma**
氣喘

**B** 兒童脆弱性
**1** 2 3 4 5 6 7
**children are not little adults**
兒童不是縮小版的大人
全場核心句

## A 機構、法規與文件
INSTITUTIONS & INSTRUMENTS

10 TERMS

| 原文 | 建議譯法 | 說明 |
|---|---|---|
| **EPA (Environmental Protection Agency)** | 美國環保署 | |
| **Office of Children's Health Protection** | 兒童健康保護辦公室 | EPA 內單位，講者時任主任 |
| **G8** | 八大工業國（G8） | 1997 邁阿密環境部長宣言 |
| **WHO** | 世界衛生組織 | 台灣慣稱世衛 |
| **Clean Air Act** | 《清淨空氣法》 | 美國 1970；台灣自己的是《空氣污染防制法》 |
| **Clean Air Act Amendments of 1990** | 1990 年《清淨空氣法》修正案 | |

| 原文 | 建議譯法 | 說明 |
|---|---|---|
| **Framework Convention on Tobacco Control (FCTC)** | 《菸草控制框架公約》 | 2003 年通過 |
| **Partnership to Eliminate Lead Paint** | 全球消除含鉛油漆聯盟 | 2010 年成立（UNEP／WHO） |
| **Partnership for Clean Fuels and Vehicles** | 潔淨燃料與車輛夥伴關係 | UNEP 主導 |
| **Pediatric Environmental Health** | 《兒童環境健康》（Pediatric Environmental Health） | 美國兒科醫學會手冊，講者主編，第 3 版有中譯 |



## B 兒童脆弱性
CHILDREN'S VULNERABILITY　　12 TERMS

| 原文 | 建議譯法 | 說明 |
|---|---|---|
| **children are not little adults** | 兒童不是縮小版的大人 | 全場核心句 |
| **hand-to-mouth activity** | 手口行為 | |
| **metabolic rate** | 代謝率 | |
| **skin surface area** | 體表面積 | |
| **excretion** | 排泄 | altered excretion = 排泄功能不同 |
| **shelf life** | 「保存期限」（餘命長） | 講者幽默說法，可譯「未來的日子還長」 |
| **organ development** | 器官發育 | |
| **kilogram for kilogram** | 以每公斤體重計 | per kilogram of body weight 同 |
| **detoxify** | 解毒 | |
| **critical windows of vulnerability** | 脆弱關鍵期 | |
| **latency period** | 潛伏期 | |
| **politically powerless** | 在政治上沒有發言權 | |

## C 疾病與負擔
DISEASE & BURDEN　　16 TERMS

| 原文 | 建議譯法 | 說明 |
|---|---|---|
| **environmental burden of disease** | 環境疾病負擔 | |
| **attributable to the environment** | 歸因於環境 | |
| **chronic diseases** | 慢性病 | |

| 原文 | 建議譯法 | 說明 |
|---|---|---|
| **developmental disabilities** | 發展障礙 | |
| **environmental degradation** | 環境惡化 | |
| **desertification** | 沙漠化 | |
| **deforestation** | 森林砍伐 | |
| **obesity** | 肥胖 | |
| **respiratory infections** | 呼吸道感染 | |
| **pneumonia** | 肺炎 | |
| **asthma** | 氣喘 | |
| **ischemic heart disease** | 缺血性心臟病 | |
| **low birthweight** | 低出生體重 | |
| **preterm births** | 早產 | |
| **IUGR (intrauterine growth restriction)** | 子宮內生長遲滯 | |
| **mental retardation** | 智能障礙 | 投影片舊用語；中文用現行「智能障礙」 |



## D 汙染源與介入
HAZARDS & INTERVENTIONS

15 TERMS

| 原文 | 建議譯法 | 說明 |
|---|---|---|
| **outdoor air pollution** | 戶外空氣汙染 | |
| **household air pollution** | 家戶空氣汙染 | indoor smoke = 室內燃燒煙霧 |
| **biomass fuel** | 生質燃料 | 柴、炭、糞便、作物殘渣 |
| **cleaner-burning stoves** | 清潔爐具／改良爐灶 | |
| **Air Quality Index (AQI)** | 空氣品質指標（**AQI**） | 台灣同用 AQI |
| **particulate matter (PM)** | 懸浮微粒 | |
| **ground-level ozone** | 地面臭氧 | |
| **secondhand smoke** | 二手菸 | |
| **cotinine** | 可丁寧 | 尼古丁代謝物；投影片備註「古丁尼」是誤植 |
| **clean indoor air laws** | 室內禁菸法／無菸法 | |
| **multi-unit housing** | 集合住宅 | |
| **subsidized housing** | 政府補貼住宅 | 台灣類似「社會住宅」 |
| **leaded gasoline** | 含鉛汽油 | |

| 原文 | 建議譯法 | 說明 |
|---|---|---|
| **lead paint** | 含鉛油漆 | |
| **third-party certification** | 第三方認證 | |

## E 預防層級
LEVELS OF PREVENTION

6 TERMS

| 原文 | 建議譯法 | 說明 |
|---|---|---|
| **primordial prevention** | 根本預防 | 從源頭不讓危險因子出現，如禁用含鉛汽油 |
| **primary prevention** | 初級預防 | 防止疾病發生 |
| **secondary prevention** | 次級預防 | 早期篩檢，如血鉛篩檢 |
| **tertiary prevention** | 三級預防 | 疾病治療，如螯合治療 |
| **effective interventions** | 有效介入措施 | |
| **return on investment** | 投資報酬 | |



## F 投影片陷阱 → 口譯說法
SLIDE TRAPS

4 PAIRS

> 別照投影片唸
> 投影片有錯字與舊用語，照意思譯。

| 投影片原文 | | 口譯說法 | |
|---|---|---|---|
| Clear Air Act | （錯字） | **Clean Air Act** | 《清淨空氣法》 |
| mental retardation | 智力遲緩（舊） | **intellectual disability** | 智能障礙 |
| 古丁尼 | （誤植） | **cotinine** | 可丁寧 |
| death per year | （文法） | **deaths per year** | 每年死亡人數 |

## G 幽默與文化梗
HUMOR & CULTURE

4 ITEMS

| | |
|---|---|
| *long "shelf life"* | 「保存期限很長」＋白話：未來的日子還長 |
| *Smoker = Glamor / Villain* | 1944 鮑嘉與白考兒；1969 庫伊拉（《101 忠狗》反派） |
| *Children are not little adults* | 「兒童不是縮小版的大人」，台灣兒科常用語 |
| *Living Worlds Apart* | 「天差地遠」 |

# 串連文章（英文）

PASSAGE FOR SIGHT TRANSLATION

03

575 words / 朗讀約 4-5 分鐘

術語以粗體金底線標示。左欄是每段的朗讀時間碼：先計時視譯一遍，再進入 04 對照。

## *Effective Interventions That Prevent Disease*

575 words / 7 段



**01**
0:00
0:34

In 1997, the leaders of the **G8** stated that protecting the health of our children is a shared fundamental value. They recognized that children face particular vulnerabilities to environmental threats. Today I want to show you that children are different from adults, and to identify successes and challenges in reducing environmental hazards. For centuries, art showed children as small adults, as in this carving from Egypt 3,300 years ago. But **children are not little adults**.

**02**
0:34
1:14

Children have **hand-to-mouth activity**, they are closer to the ground, and they take in more air and food. They have a higher **metabolic rate**, more **skin surface area**, different **excretion**, ongoing **organ development**, and a long **shelf life**. A developing child's chemical exposures are greater **kilogram for kilogram** than an adult's. The systems that **detoxify** and excrete industrial chemicals are not fully developed. And children live in worlds apart: in Philadelphia, lead in tap water; in the Philippines, no tap at all, and mercury from small-scale gold mining.

**03**
1:14
1:42

Child **chronic diseases** and **developmental disabilities** are increasing. Industrialization, urbanization, poverty, climate change, **desertification** and **deforestation** all drive **environmental degradation**, and children suffer most of the effects. In the USA, the prevalence of **obesity** in children has nearly quadrupled. Each year, 1.7 million deaths of children under 5 are **attributable to the environment**. Of these, 570,000 are from **respiratory infections**, including **pneumonia**.

**04**
1:42
2:20

**Outdoor air pollution** from industry and vehicles is clearly related to respiratory disease, especially **asthma**. The **Clean Air Act** of 1970 defines EPA's responsibility for protecting the nation's air quality. At the individual level, people can check the **Air Quality Index (AQI)** for ozone, **particulate matter (PM)**, carbon

monoxide, sulfur dioxide and nitrogen dioxide. The benefits of the 1990 amendments will reach about $2 trillion in 2020, against costs of about $65 billion. Over 230,000 early deaths are prevented, and the benefits exceed the costs by more than 30 to 1.

**05**
2:20
2:49

Exposure to **biomass fuel** causes **household air pollution**. Children inhale more pollutants per kilogram of body weight, and because their airways are narrower, irritation causes proportionately greater obstruction. The good news is that **cleaner-burning stoves** and cleaner fuels such as gas and biogas reduce indoor smoke. A switch to cleaner fuels gives a 7-fold **return on investment**, and improved stoves a 25- to 60-fold return.

**06**
2:49
3:40

**Secondhand smoke** causes more than 600,000 deaths per year, more than 1% of all deaths. Smokers put 1.8 billion non-smokers at risk. Most tobacco use starts before the age of 18, and a quarter of it before 10. If current trends continue, 250 million children and teenagers alive today will die from tobacco-related diseases. In 1944 the smoker was glamorous; by 1969 the smoker was a villain, and in 2003, 192 WHO member states adopted the **Framework Convention on Tobacco Control (FCTC)**. Smoke-free laws work, yet only 18% of the world lives under them. In Scotland, **clean indoor air laws** reduced **preterm births** and hospital admissions for childhood asthma. In US **multi-unit housing**, banning smoking could save $521 million a year.

**07**
3:40
4:25

Lead exposure accounts for about 1% of the global burden of disease, and in developing countries as much as 15 to 20% of **mental retardation** could be caused by lead. Unrelenting advocacy led to the ban on **leaded gasoline** in the US in the 1970s, and children's blood lead levels dropped. That is **primordial prevention**: removing the hazard before it ever reaches the child. **Lead in paint** is the unfinished agenda; the **Partnership to Eliminate Lead Paint**, launched in 2010, promotes **third-party certification**. We can make a difference: become more aware, move towards primordial prevention, and use the **effective interventions** we already have.

# 中英對照解析

SENTENCE-ALIGNED ANALYSIS

04

37 句 / 46 個切分單位

一列一個切分單位：左欄講者原文，右欄照原文順序的參考口譯，由上往下讀就是口譯廂裡的順序。每段後的解析分知識點、口譯技巧、投影片、數字預讀。

## 01 開場：兒童不是縮小版的大人

兒童環境脆弱性是國際共識

0:00–0:34
5 句

| | SOURCE | 參考口譯（照原文順序） |
|---|---|---|
| 1.1 | In 1997, the leaders of the **G8** stated that protecting the health of our children is a shared fundamental value. | 在 1997 年，八大工業國領袖宣布：保護兒童健康，是我們共同的基本價值。 |
| 1.2 | They recognized that children face particular vulnerabilities to environmental threats. | 他們也承認，兒童面對環境威脅時特別脆弱。 |
| 1.3 | Today I want to show you that children are different from adults, | 今天我想說明，兒童和成人不一樣， |
| | and to identify successes and challenges | 也要談談成功經驗與挑戰， |
| | in reducing environmental hazards. | 看如何減少環境危害。 |
| 1.4 | For centuries, art showed children as small adults, as in this carving from Egypt 3,300 years ago. | 好幾百年來，藝術都把兒童畫成小大人，像這件 3,300 年前埃及的浮雕。 |
| 1.5 | But **children are not little adults**. | 但是，兒童不是縮小版的大人。 |

投影片　第 5–6 張是埃及阿肯那頓與娜芙蒂蒂和三個女兒的浮雕，女兒畫得像縮小的成人。

口譯技巧　核心句先定譯，後面再出現直接重複，不要換說法。

## 02 兒童為什麼不同

暴露多、器官未熟、解毒弱、餘命長

0:34–1:14
5 句

| | SOURCE | 參考口譯（照原文順序） |
|---|---|---|
| 2.1 | Children have **hand-to-mouth activity**, they are closer to the ground, and they take in more air and food. | 兒童有手口行為，離地面比較近，吸入的空氣、吃下的食物也比較多。 |
| 2.2 | They have a higher **metabolic rate**, more **skin surface area**, different **excretion**, ongoing **organ development**, and a long **shelf life**. | 他們的代謝率較高，體表面積相對較大，排泄功能不同，器官發育還在進行，而且保存期限很長，也就是未來的日子還很長。 |

| | | |
|---|---|---|
| 2.3 | A developing child's chemical exposures are greater **kilogram for kilogram** than an adult's. | 發育中兒童的化學暴露，以每公斤體重計比成人更多。 |
| 2.4 | The systems that **detoxify** and excrete industrial chemicals are not fully developed. | 負責解毒和排出工業化學物質的系統，還沒發育完全。 |
| 2.5 | And children live in worlds apart: in Philadelphia, lead in tap water; in the Philippines, no tap at all, and mercury from small-scale gold mining. | 而且兒童的處境天差地遠：在費城，是自來水裡有鉛；在菲律賓，根本沒有自來水，還有小規模採金帶來的汞。 |

投影片　第 9 張左右對照：費城 = 自來水含鉛、食物殘留農藥、球鞋含汞；菲律賓 = 沒有自來水、誤喝農藥致死、採金汞汙染。

口譯技巧　shelf life 是幽默梗：先直譯「保存期限」再補一句白話，觀眾才笑得出來。

## 03 疾病負擔正在上升

每年 170 萬五歲以下死亡歸因環境

1:14–1:42
5 句

| | SOURCE | 參考口譯（照原文順序） |
|---|---|---|
| 3.1 | Child **chronic diseases** and **developmental disabilities** are increasing. | 兒童慢性病和發展障礙正在增加。 |
| 3.2 | Industrialization, urbanization, poverty, climate change, **desertification** and **deforestation** all drive **environmental degradation**, and children suffer most of the effects. | 工業化、都市化、貧窮、氣候變遷、沙漠化和森林砍伐都在推動環境惡化，而受害最深的是兒童。 |
| 3.3 | In the USA, the prevalence of **obesity** in children has nearly quadrupled. | 在美國，兒童肥胖盛行率將近翻了四倍。 |
| 3.4 | Each year, 1.7 million deaths of children under 5 are **attributable to the environment**. | 每年有 170 萬名 5 歲以下兒童的死亡，歸因於環境。 |
| 3.5 | Of these, 570,000 are from **respiratory infections**, including **pneumonia**. | 其中 57 萬名死於呼吸道感染，包括肺炎。 |

數字預讀　WHO 2017：170 萬總數 | 呼吸道感染 57 萬 | 腹瀉 36.1 萬 | 新生兒疾病 27 萬 | 瘧疾 20 萬 | 非蓄意傷害 20 萬；圖上 26%。

知識點　WHO 環境定義「包含」空水土汙染、輻射、噪音、職業、建成環境、氣候、洗手；「不含」菸酒毒、飲食、蚊帳、失業、天災、人傳人。

## 04 戶外空汙：法規見效

《清淨空氣法》效益超過成本 30 倍

1:42–2:20
5 句

| | SOURCE | 參考口譯（照原文順序） |
|---|---|---|
| 4.1 | **Outdoor air pollution** from industry and vehicles is clearly related to respiratory disease, especially **asthma**. | 工業和車輛造成的戶外空氣汙染，明顯和呼吸道疾病有關，尤其是氣喘。 |

| | | |
|---|---|---|
| 4.2 | The **Clean Air Act** of 1970 defines EPA's responsibility for protecting the nation's air quality. | 1970 年的《清淨空氣法》，規定了美國環保署保護全國空氣品質的責任。 |
| 4.3 | At the individual level, people can check the **Air Quality Index (AQI)** for ozone, **particulate matter (PM)**, carbon monoxide, sulfur dioxide and nitrogen dioxide. | 在個人層面，民眾可以查看臭氧、懸浮微粒、一氧化碳、二氧化硫和二氧化氮的空氣品質指標（AQI）。 |
| 4.4 | The benefits of the 1990 amendments will reach about $2 trillion in 2020, | 1990 年修正案的效益，到 2020 年會達到約 2 兆美元， |
| | against costs of about $65 billion. | 成本則約 650 億美元。 |
| 4.5 | Over 230,000 early deaths are prevented, | 超過 23 萬人免於早逝， |
| | and the benefits exceed the costs by more than 30 to 1. | 效益超過成本 30 比 1 以上。 |

投影片　第 27 張有錯字 Clear Air Act，照 Clean 譯。

知識點　數字來自 EPA 2011 年前瞻研究的「預估」；2017 年講時仍是未來式 will reach。見背景解說 2。

口譯技巧　兆／億換算先備好：2 trillion = 2 兆；65 billion = 650 億。



## 05 家戶空汙：改良爐灶

兒童氣道窄，換乾淨爐灶報酬高

2:20–2:49
4 句

| | SOURCE | 參考口譯（照原文順序） |
|---|---|---|
| 5.1 | Exposure to **biomass fuel** causes **household air pollution**. | 接觸生質燃料，會造成家戶空氣汙染。 |
| 5.2 | Children inhale more pollutants per kilogram of body weight, | 兒童以每公斤體重計吸入更多汙染物， |
| | and because their airways are narrower, | 而且因為氣道比較窄， |
| | irritation causes proportionately greater obstruction. | 受到刺激時，阻塞程度相對更嚴重。 |
| 5.3 | The good news is that **cleaner-burning stoves** and cleaner fuels such as gas and biogas reduce indoor smoke. | 好消息是，清潔爐具和瓦斯、沼氣等乾淨燃料，可以減少室內燃燒煙霧。 |
| 5.4 | A switch to cleaner fuels gives a 7-fold **return on investment**, | 改用乾淨燃料，投資報酬是 7 倍； |
| | and improved stoves a 25- to 60-fold return. | 改良爐灶更有 25 到 60 倍。 |

投影片　第 30 張照片：奈及利亞母親背著孩子生火（2008）；風險：急性上呼吸道疾病、氣喘、肺炎、嬰兒早夭。

口譯技巧　Reason for Hope 是講者反覆出現的轉折詞，固定譯「希望所在」。

## 06 菸害：從魅力到反派

禁菸法規有效，但只涵蓋 18% 人口

2:49–3:40
8 句

| | SOURCE | 參考口譯（照原文順序） |
|---|---|---|
| 6.1 | **Secondhand smoke** causes more than 600,000 deaths per year, more than 1% of all deaths. | 二手菸每年造成超過 60 萬人死亡，占所有死亡的 1% 以上。 |
| 6.2 | Smokers put 1.8 billion non-smokers at risk. | 吸菸者讓 18 億不吸菸者暴露在風險中。 |
| 6.3 | Most tobacco use starts before the age of 18, and a quarter of it before 10. | 大多數人在 18 歲前開始吸菸，其中四分之一更在 10 歲前。 |
| 6.4 | If current trends continue, 250 million children and teenagers alive today will die from tobacco-related diseases. | 照目前趨勢，今天活著的兒童和青少年中，將有 2.5 億人死於菸草相關疾病。 |
| 6.5 | In 1944 the smoker was glamorous; by 1969 the smoker was a villain, and in 2003, 192 WHO member states adopted the **Framework Convention on Tobacco Control (FCTC)**. | 在 1944 年，吸菸者代表魅力；到了 1969 年，吸菸者成了反派；在 2003 年，192 個世衛會員國通過了《菸草控制框架公約》。 |
| 6.6 | Smoke-free laws work, yet only 18% of the world lives under them. | 禁菸法規有效，但全世界只有 18% 的人口受到保護。 |
| 6.7 | In Scotland, **clean indoor air laws** reduced **preterm births** and hospital admissions for childhood asthma. | 在蘇格蘭，室內禁菸法減少了早產，也減少兒童氣喘住院。 |
| 6.8 | In US **multi-unit housing**, banning smoking could save $521 million a year. | 在美國的集合住宅，全面禁菸每年可省下 5.21 億美元。 |

數字預讀　2004 二手菸死亡：缺血性心臟病 37.9 萬｜下呼吸道感染 16.5 萬｜氣喘 3.69 萬｜肺癌 2.14 萬。集合住宅：補貼住宅吸菸率 32.7% vs 全美成人 20.6%；5.21 億＝醫療 3.41 億＋翻修 1.08 億＋火災 7,200 萬。

投影片　1944 是亨佛萊鮑嘉與洛琳白考兒（Bogart／Bacall），1969 是《101 忠狗》庫伊拉（Cruella）。It took 60 years＝「花了 60 年」。第 41 張備註「古丁尼」應為可丁寧（cotinine）。

## 07 鉛：汽油成功，油漆未竟

禁用含鉛汽油使血鉛下降，油漆仍待努力

3:40–4:25
5 句

| | SOURCE | 參考口譯（照原文順序） |
|---|---|---|
| 7.1 | Lead exposure accounts for about 1% of the global burden of disease, | 鉛暴露約占全球疾病負擔的 1%， |
| | and in developing countries | 而在開發中國家， |
| | as much as 15 to 20% of **mental retardation** could be caused by lead. | 多達 15% 到 20% 的智能障礙可能是鉛造成的。 |
| 7.2 | Unrelenting advocacy led to the ban on **leaded gasoline** in the US in the 1970s, and children's blood lead levels dropped. | 持續不懈的倡議，促成美國在 1970 年代禁用含鉛汽油，兒童血鉛也隨之下降。 |

| | | |
|---|---|---|
| 7.3 | That is **primordial prevention**: removing the hazard before it ever reaches the child. | 這就是根本預防：在危害接觸到孩子之前，就從源頭移除。 |
| 7.4 | **Lead in paint** is the unfinished agenda; the **Partnership to Eliminate Lead Paint**, launched in 2010, promotes **third-party certification**. | 含鉛油漆則是尚未完成的工作；2010 年成立的全球消除含鉛油漆聯盟，推動第三方認證。 |
| 7.5 | We can make a difference: become more aware, move towards primordial prevention, and use the **effective interventions** we already have. | 我們可以帶來改變：提高警覺，邁向根本預防，善用已經有的有效介入措施。 |

知識點 投影片「截至 2016 年 6 月」仍有少數國家用含鉛汽油；阿爾及利亞 2021 年 7 月停售後，UNEP 宣布全球已全面淘汰。講者照舊說就照譯，必要時心裡有數。見背景解說 3。

知識點 mental retardation 為舊稱，中文用「智能障礙」即可。

口譯技巧 預防金字塔四層：根本／初級／次級／三級，聽到 primordial 不要譯成「初級」。

# 口譯廂速查卡

BOOTH CARD

05

22 組 / 印出來放在口譯廂

| | |
|---|---|
| **children are not little adults** | 兒童不是縮小版的大人 |
| **primordial prevention** | 根本預防 |
| **primary prevention** | 初級預防 |
| **secondary prevention** | 次級預防 |
| **tertiary prevention** | 三級預防 |
| **Clean Air Act** | 《清淨空氣法》 |
| **Air Quality Index (AQI)** | 空氣品質指標（AQI） |
| **particulate matter (PM)** | 懸浮微粒 |
| **household air pollution** | 家戶空氣汙染 |
| **biomass fuel** | 生質燃料 |
| **secondhand smoke** | 二手菸 |
| **Framework Convention on Tobacco Control (FCTC)** | 《菸草控制框架公約》 |
| **clean indoor air laws** | 室內禁菸法／無菸法 |
| **leaded gasoline** | 含鉛汽油 |
| **lead paint** | 含鉛油漆 |
| **environmental burden of disease attributable to the environment** | 環境疾病負擔歸因於環境 |
| **hand-to-mouth activity** | 手口行為 |
| **low birthweight** | 低出生體重 |
| **IUGR (intrauterine growth restriction)** | 子宮內生長遲滯 |
| **return on investment** | 投資報酬 |
| **effective interventions** | 有效介入措施 |



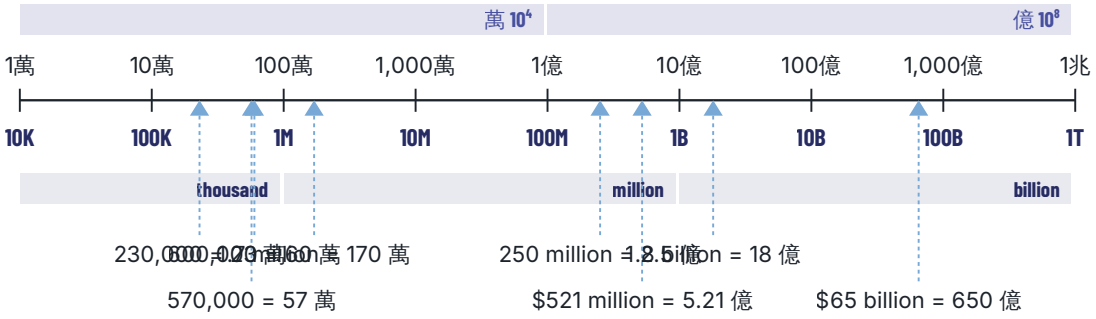
## 臨場原則

**01** primordial = 根本預防，primary = 初級預防，全場不可混。

**02** Reason for Hope 固定譯「希望所在」；講者每段都用它轉折。

**03** 含鉛汽油已於 2021 年全球淘汰；講者若說「仍有國家使用」照譯，不糾正。

## 數字刻度尺　中文四位一組，英文三位一組




**$2 trillion** = 2 兆 | **a quarter** = 四分之一（25%） | **1970s** = 1,970 | **30%** = 三成 | **one in five** = 五分之一

民國年 **+ 1911** = 西元年